This is going to be a loosely held-together collection of notes on different types of constructions in the field of operator algebras whose primary connection is that they're referred to by "product."

Tensor Products of C^*-Algebras

One of the main complications with C^*-algebras is that the algebraic tensor product $A \odot B$ may admit more than one C^*-norm.

> **Definition:** A C^*-norm on $A \odot B$ is a norm $\|\cdot\|_\alpha$ satisfying $\|xy\|_\alpha \leq \|x\|_\alpha \|y\|_\alpha$ and $\|x^*x\|_\alpha = \|x\|_\alpha^2$ for all $x, y \in A \odot B$. The completion of $A \odot B$ with respect to this norm is denoted $A \otimes_\alpha B$.

However, the existence of there being multiple norms is not necessarily universal.

For instance, since there is a $*$-isomorphism between $\mathbb{M}_n(\mathbb{C}) \odot A$ and $\mathbb{M}_n(A)$, we have that $\mathbb{M}_n(\mathbb{C}) \odot A$ admits a unique C^*-norm, which is the norm that it obtains from $\mathbb{M}_n(A)$.

Two C^*-Norms on the Tensor Product

> **Definition:** Let A and B are C^*-algebras. The *maximal norm* on $A \odot B$ is given by
>
> $$\|x\|_{\text{max}} = \sup\{\|\pi(x)\| : \pi \colon A \odot B \to B(H) \text{ a cyclic } *\text{-representation}\}$$
>
> for any $x \in A \odot B$. The completion of $A \odot B$ with respect to $\|\cdot\|_{\text{max}}$ is denoted $A \otimes_{\text{max}} B$.

This is akin to the universal C^*-norm on the group C^*-algebra $C^*(\Gamma) \subseteq B(\ell_2(\Gamma))$.

> **Definition:** If $\pi \colon A \to B(H)$ and $\sigma \colon B \to B(K)$ are faithful representations, then the *spatial* or *minimal* C^*-norm on $A \odot B$ is given by
>
> $$\left\| \sum_{i=1}^{n} a_i \otimes b_i \right\|_{\text{min}} = \left\| \sum_{i=1}^{n} \pi(a_i) \otimes \sigma(b_i) \right\|_{B(H \otimes K)}.$$
>
> The completion of $A \odot B$ with respect to $\|\cdot\|_{\text{min}}$ is denoted $A \otimes_{\text{min}} B$.

> **Remark:** If we consider von Neumann algebras $M \subseteq B(H)$ and $N \subseteq B(K)$, then there are numerous C^*-norms that $M \odot N$ can be endowed with, but their norm completions will not be von Neumann algebras. The von Neumann algebra tensor product is still well-defined, and denoted $M \bar{\otimes} N$, which is the von Neumann algebra generated by $M \otimes N \subseteq B(H \otimes K)$. There are some more structural properties of tensor products of von Neumann algebras that inform us about their representation theory, which will be discussed in the next section.

There are a variety of technicalities we must consider before moving forward. First, we must verify that $\|\cdot\|_{\text{max}}$ is finite; for this, observe that if $\pi \colon A \odot B \to B(H)$ is any representation, then we may restrict to π_A and π_B, such that

$$\begin{aligned} \|\pi(a \otimes b)\| &\leq \|\pi_A(a)\| \|\pi_B(b)\| \\ &\leq \|a\| \|b\|, \end{aligned}$$

meaning $\|x\|_{\text{max}} < \infty$ for all $x \in A \odot B$.

The other major questions we must consider are the following.

(i) Are $\|\cdot\|_{\text{max}}$ and $\|\cdot\|_{\text{min}}$ even C^*-norms?

(ii) Is $\|\cdot\|_{\text{min}}$ independent of the choice of faithful representation?

(iii) Is it possible to reduce the non-unital case to the unital case?

We can dispense with (iii) by observing that, in fact, any C^*-norm on the tensor product $A_1 \odot A_2$ can be uniquely extended to the tensor product of their unitizations $\widetilde{A_1} \odot \widetilde{A_2}$.

Proposition: Let A_1 and A_2 be non-unital C^*-algebras with respective unitizations $\widetilde{A_1}$ and $\widetilde{A_2}$. Then, if $\|\cdot\|_\beta$ is any C^*-norm on the tensor product $A_1 \odot A_2$, it can be extended to a C^*-norm on $\widetilde{A_1} \odot \widetilde{A_2}$.

Proof. Denote by $A_\beta = A_1 \otimes_\beta A_2$. Let $\pi \colon A_\beta \to B(H)$ be a faithful representation, meaning that $\|x\|_\beta = \|\pi(x)\|$ for all $x \in A_1 \odot A_2$.

If π_1 and π_2 are the respective restrictions of π to A_1 and A_2, then we may extend π_1 and π_2 to the unitizations $\widetilde{A_1}$ and $\widetilde{A_2}$ respectively by taking

$$\pi_i^0(x + \lambda 1) = \pi_i(x) + \lambda 1$$

for each $x \in A_i$ and $\lambda \in \mathbb{C}$. The ranges of π_1^0 and π_2^0 commute, meaning that we may define a representation π_0 on $\widetilde{A_1} \odot \widetilde{A_2}$ by

$$\pi_0(x_1 \otimes x_2) = \pi_1^0(x_1)\pi_2^0(x_2).$$

Observe that π_0 extends π. This allows us to define a C^*-norm β_0 on $\widetilde{A_1} \odot \widetilde{A_2}$ by taking $\|x\|_{\beta_0} = \|\pi_0(x)\|$. This is our desired extension to the unitization. □

Tackling questions (i) and (ii) is more difficult. For this, consider the fact that $\|\cdot\|_{\max}$ admits a universal property as described below, which follows from the fact that any C^*-algebra can be faithfully represented on some Hilbert space.

Proposition: Let $\pi \colon A \odot B \to C$ be a $*$-homomorphism. Then, there exists a unique $*$-homomorphism $A \otimes_{\max} B \to C$ that extends π. In particular, a pair of $*$-homomorphisms $\pi_A \colon A \to C$ and $\pi_B \colon B \to C$ with commuting ranges induces a unique $*$-homomorphism on $A \otimes_{\max} B$.

Corollary: The norm $\|\cdot\|_{\max}$ is the largest possible C^*-norm on $A \odot B$.

Proof. If $\|\cdot\|_\alpha$ is any other C^*-norm, then there is a surjective $*$-homomorphism $A \otimes_{\max} B \to A \otimes_\alpha B$ (extend the identity map), which means that $\|x\|_\alpha \leq \|x\|_{\max}$ for all $x \in A \odot B$. □

In particular, this means that $\|\cdot\|_{\max}$ dominates $\|\cdot\|_{\min}$, so we only need to show that $\|\cdot\|_{\min}$ is a norm.

Lemma: The product $*$-homomorphism $B(\mathcal{H}) \odot B(\mathcal{K}) \to B(\mathcal{H} \otimes \mathcal{K})$ induced by the representations $B(\mathcal{H}) \cong B(\mathcal{H}) \otimes \mathbb{C}1_{\mathcal{K}}$ and $B(\mathcal{K}) \cong \mathbb{C}1_{\mathcal{H}} \otimes B(\mathcal{K})$ is injective.

Proof. We will show that if a finite sum of tensor product operators $\sum_i S_i \otimes T_i \in B(\mathcal{H} \otimes \mathcal{K})$ is zero, then the corresponding sum of elementary tensors is also zero. We will assume that the operators S_i are linearly independent.

For this, we have for all $v, w \in \mathcal{H}$ and $\xi, \eta \in \mathcal{K}$, that

$$\begin{aligned}
\left\langle \left(\sum_i S_i \otimes T_i \right)(v \otimes \xi), w \otimes \eta \right\rangle = 0 \\
&= \sum_i \langle (S_i \otimes T_i)(v \otimes \xi), w \otimes \eta \rangle \\
&= \sum_i \langle S_i v, w \rangle \langle T_i \xi, \eta \rangle \\
&= \left\langle \left(\left(\sum_i \langle T_i \xi, \eta \rangle \right) S_i \right) v, w \right\rangle.
\end{aligned}$$

Since this holds for all $v, w \in \mathcal{H}$, we have that the operator $\sum_i \langle T_i \xi, \eta \rangle S_i = 0$, so by linear independence, each of the $\langle T_i \xi, \eta \rangle = 0$, meaning that $T_i = 0$ for each i. □

Corollary: The quantity $\|\cdot\|_{\min}$ is a norm.

Proposition: The spatial tensor product is independent of the choices of faithful representation $\pi \colon A \to B(\mathcal{H})$ and $\sigma \colon B \to B(\mathcal{K})$.

Proof. We will let $\|\cdot\|_{(\pi,\sigma)}$ denote the minimal norm with respect to π and σ. It suffices to prove that if $\sigma'\colon B \to B(\mathcal{K}')$ is another faithful representation, then $\|\cdot\|_{(\pi,\sigma)} = \|\cdot\|_{(\pi,\sigma')}$.

Additionally, we will assume we are in the separable setting. Let $(P_n)_n$ be an increasing net of finite-rank projections in $B(\mathcal{H})$ with P_n having rank n, and $\|P_n h - h\| \to 0$ for all $h \in \mathcal{H}$. It follows that for all $X \in B(\mathcal{H}\otimes\mathcal{K})$, we have

$$\|X\| = \sup_n \|(P_n \otimes 1_{\mathcal{K}})X(P_n \otimes 1_{\mathcal{K}})\|.$$

If $\sum_{i=1}^k a_i \otimes b_i \in A \odot B$ is arbitrary, then

$$\begin{aligned}\left\|\sum_{i=1}^k a_i \otimes b_i\right\|_{(\pi,\sigma)} &= \sup_n \left\|\sum_{i=1}^k (P_n \pi(a_i) P_n) \otimes \sigma(b_i)\right\|\\ \left\|\sum_{i=1}^k a_i \otimes b_i\right\|_{(\pi,\sigma')} &= \sup_n \left\|\sum_{i=1}^k (P_n \pi(a_i) P_n) \otimes \sigma'(b_i)\right\|.\end{aligned}$$

Yet, since $P_n B(\mathcal{H}) P_n$ is isomorphic to $\mathbb{M}_n(\mathbb{C})$, we have

$$\left\|\sum_{i=1}^k (P_n \pi(a_i) P_n) \otimes \sigma(b_i)\right\| = \left\|\sum_{i=1}^k (P_n \pi(a_i) P_n) \otimes \sigma'(b_i)\right\|$$

for each n, since there is a unique norm on $\mathbb{M}_n(\mathbb{C}) \odot B$. □

Takesaki's Theorem

Now we will show that $\|\cdot\|_{\min}$ is actually the smallest C^*-norm on $A \odot B$. We will show that for any pair of linear functionals $\varphi \in A^*$ and $\psi \in B^*$, and any C^*-norm $\|\cdot\|_\alpha$ on $A \odot B$, then the algebraic tensor product $\varphi \odot \psi$ is automatically continuous with respect to $\|\cdot\|_\alpha$.

Lemma: Let $S(A)$ denote the state space of a C^*-algebra A. If $V \subseteq S(A)$ is a set of states such that for every self-adjoint $a \in A_{\text{s.a.}}$, we have

$$\|a\| = \sup_{\varphi \in U} \left|\varphi(a)\right|,$$

then $S(A) \subseteq \overline{\operatorname{conv}}^{w^*}(V)$.

Proof. Let $U = \overline{\operatorname{conv}}^{w^*}(V)$. Suppose there is $\psi \in S(A)$ that is not in U. By Hahn–Banach separation, there is $a \in A$ and $t \in \mathbb{R}$ such that $\operatorname{Re}(\varphi(a)) < t < \operatorname{Re}(\psi(a))$ for all $\varphi \in U$. Replacing $a = \operatorname{Re}(a)$, we may assume that a is self-adjoint. If a were positive, then we would have our contradiction as $\psi(a) \leq \|a\|$.

If A is unital, then $a + \|a\|1 \geq 0$, so that

$$\begin{aligned} 0 \leq \varphi(a + \|a\|1) &= \varphi(a) + \|a\|\\ &< t + \|a\|\\ &< \psi(a + \|a\|1), \end{aligned}$$

so that

$$\begin{aligned} \|a + \|a\|1\| &= \sup_{\varphi \in V} \left|\varphi(a + \|a\|1)\right|\\ &\leq t + \|a\|\\ &< \psi(a + \|a\|1)\\ &\leq \|a + \|a\|1\|, \end{aligned}$$

which gives our desired contradiction. If A is non-unital, then we pass to the unitization and apply the same argument. $\square$

Next, we try to understand when certain states on the tensor product can be decomposed into a product of states.

Lemma: Let $A \subseteq C$, $B = A' \cap C$ its relative commutant, and ξ a state on C such that $\xi|_A$ is pure. Then, $\xi(ab) = \xi(a)\xi(b)$ for all $a \in A$ and $b \in B$.

Proof. We may assume that $b \in B$ is a positive element of norm 1.

First, assume $\xi(b) = 0$. Then, for all $a \in A$, Cauchy–Schwarz gives

$$\begin{aligned} |\xi(ab)|^2 &= \left|\xi\left(\left(ab^{1/2}\right)\left(b^{1/2}\right)\right)\right|^2 \\ &\leq \xi(aba^*)\xi(b), \end{aligned}$$

so $\xi(ab) = 0 = \xi(a)\xi(b)$.

Now, assume $\xi(b) = 1$ Then, passing to unitizations if necessary, we have $\xi(1-b) = 0$, so we have

$$\begin{aligned} 0 &= \xi(a(1-b)) \\ &= \xi(a) - \xi(ab), \end{aligned}$$

so $\xi(ab) = \xi(a) = \xi(a)\xi(b)$.

Finally, if $0 < \xi(b) < 1$, then

$$\xi(a) = \xi(b)\left(\frac{1}{\xi(b)}\xi(ab)\right) + (1-\xi(b))\left(\frac{1}{1-\xi(b)}\xi(a(1-b))\right),$$

so we see that $\xi|_A$ is then a convex combination of states on A, which means that $\xi(a) = \frac{1}{\xi(b)}\xi(ab)$, and we are done. $\square$

If $\|\cdot\|_\alpha$ is a C^*-norm on $A \odot B$, with completion $A \otimes_\alpha B$, and ξ a state on $A \otimes_\alpha B$, we define the restrictions $\xi|_A$ and $\xi|_B$ as follows. If $\left(\pi_\xi, \mathcal{H}_\xi, w_\xi\right)$ is the GNS triplet, and $\pi_{\xi,A}, \pi_{\xi,B}$ the restriction homomorphisms to (the images of) A and B respectively, then

$$\begin{aligned} \xi|_A(a) &= \left\langle \pi_{\xi,A}(a)w_\xi, w_\xi \right\rangle \\ \xi|_B(b) &= \left\langle \pi_{\xi,B}(b)w_\xi, w_\xi \right\rangle. \end{aligned}$$

Proposition: Let $\|\cdot\|_\alpha$ be a C^*-norm on $A \odot B$, and let ξ be a state on $A \otimes_\alpha B$.

(i) If the restriction $\xi|_A$ is pure, then $\xi|_{A\odot B} = \xi|_A \odot \xi|_B$.

(ii) If A is abelian and ξ is a pure state on $A \otimes_\alpha B$, then both $\xi|_\alpha$ and $\xi|_\beta$ are pure with $\xi|_{A\odot B} = \xi|_A \odot \xi|_B$.

Proof. We start by showing (i). Suppose $\xi|_A$ is pure. Applying the lemma to the vector state $T \mapsto \left\langle Tv_\xi, v_\xi \right\rangle$ on $B\left(\mathcal{H}_\xi\right)$, we have

$$\begin{aligned} \xi(a \otimes b) &= \left\langle \pi_{\xi,A}(a)\pi_{\xi,B}(a)v_\xi, v_\xi \right\rangle \\ &= \left\langle \pi_{\xi,A}(a)v_\xi, \pi_{\xi,B}(b)\xi \right\rangle \\ &= \xi|_A(a)\xi|_B(b), \end{aligned}$$

so $\xi(x) = (\xi|_A \odot \xi|_B)(x)$ for all $x \in A \odot B$.

For (ii), assume A is abelian and ξ is a pure state on $A \otimes_\alpha B$. Then, the WOT closure of $\pi_\xi(A \otimes_\alpha B)$ is $B\left(\mathcal{H}_\xi\right)$. Since $\pi_\xi(A \otimes_\alpha B)$ is contained in the C^*-algebra generated by $\pi_{\xi,A}(A)$ and $\pi_{\xi,B}(B)$; since A is abelian, and $\pi_{\xi,A}$ and $\pi_{\xi,B}$ have commuting ranges, we have that $\pi_{\xi,A}(A)$ is contained in the center of $B\left(\mathcal{H}_\xi\right)$.

In particular, this means $\pi_{\xi,A}(A) = \mathbb{C}1$. It follows that $\xi|_A$ is a character, hence a pure state, and $\pi_{\xi,B}(B)$ is all of $B(\mathcal{H}_\xi)$. Therefore, $\xi|_B$ is pure. $\square$

Lemma: Let $\varphi \in S(A)$ and $\psi \in S(B)$. Then, $\varphi \odot \psi\colon A \odot B \to \mathbb{C}$ is algebraically positive.

Proof. Passing to GNS representations, we have a representation $\pi_\varphi \otimes \pi_\psi\colon A \otimes B \to B(\mathcal{H}_\varphi \otimes \mathcal{H}_\psi)$, given by

$$\varphi \odot \psi(x) = \left\langle \pi_\varphi \otimes \pi_\psi(x) v_\varphi \otimes v_\psi, v_\varphi \otimes v_\psi \right\rangle$$

for all $x \in A \odot B$, from which we get our desired result. $\square$

The next step is to show that the product states are always continuous on $A \otimes_\alpha B$.

Lemma: Suppose A and B are unital and abelian. Then, for any C^*-norm $\|\cdot\|_\alpha$ on $A \odot B$ and pair of pure states $\varphi \in S(A)$ and $\psi \in S(B)$, the linear functional $\varphi \odot \psi\colon A \odot B \to \mathbb{C}$ extends to a state on $A \otimes_\alpha B$.

Proof. Let $P(A)$ and $P(B)$ denote the pure states on A and B respectively. Define

$$\mathcal{U} = \left\{(\varphi, \psi) \in P(A) \times P(B) : \left|\varphi \odot \psi(x)\right| \leq \|x\|_\alpha \text{ for all } x \in A \odot B\right\}.$$

Then, we claim that $\mathcal{U} = P(A) \times P(B)$.

Suppose $\mathcal{U}$ is a proper subset of $P(A) \times P(B)$. Then, $\mathcal{U}$ is closed in the product topology, so there are open subsets $U_A \subseteq P(A)$ and $U_B \subseteq P(B)$ such that $(U_A \times U_B) \cap \mathcal{U} = \emptyset$. Representing $A = C(X)$ and $B = C(Y)$, let $0 \leq f \in A$ and $0 \leq g \in B$ be norm-one elements whose supports are contained in U_A and U_B respectively. Then, we have that $\varphi \odot \psi(f \otimes g) = 0$ for every $(\varphi, \psi) \in \mathcal{U}$.

Yet, this gives a contradiction by letting $\xi \in P(A \otimes_\alpha B)$ be a pure state with $\xi(f \otimes g) > 0$, and defining $\xi|_A$ and $\xi|_B$ as restrictions. Then, we have that ξ is an extension of $\xi|_A \odot \xi|_B$, with both $\xi|_A$ and $\xi|_B$ contained in $\mathcal{U}$. $\square$

Next, we can show the case for one tensor factor abelian.

Lemma: Let B be unital, A unital and abelian, and φ a pure state on A. Then, for every C^*-norm on $A \odot B$ and every state $\psi \in S(B)$, the linear functional $\varphi \odot \psi\colon A \odot B \to \mathbb{C}$ extends to a state on $A \otimes_\alpha B$.

Proof. Set

$$\mathcal{S}_\varphi = \left\{\psi \in S(B) : \left|\varphi \odot \psi(x)\right| \leq \|x\|_\alpha \text{ for all } x \in A \odot B\right\}.$$

We see that $\mathcal{S}_\varphi$ is convex and w^*-closed in $S(B)$. We claim that $\mathcal{S}_\varphi = S(B)$.

Let $h \in B$ be any self-adjoint element, and let $C = C^*(h, 1_B)$ the unital C^*-algebra generated by h. Let $A \otimes_\alpha C$ be the completion of $A \odot C$ with respect to the restriction of $\|\cdot\|_\alpha$. Let $\psi \in P(C)$ be a pure state with $\left|\psi(h)\right| = \|h\|$, and let $\varphi \otimes \psi\colon A \otimes_\alpha C \to \mathbb{C}$ be the extension given by the previous lemma (as C is abelian).

Then, $\varphi \otimes \psi$ is a pure state, so it extends to a state ξ on $A \otimes_\alpha B$. Since $\xi|_A = \varphi$ is pure, we have ξ is an extension of $\varphi \odot \xi|_B$, meaning that $\xi|_B \in \mathcal{S}_\varphi$, with $\|h\| = |\xi|_B(h)|$ as $\xi|_B(h) = \psi(h)$. Thus, the closed convex hull of $\mathcal{S}_\varphi$ contains $S(B)$, so $\mathcal{S}_\varphi$ is equal to $S(B)$. $\square$

Proposition: If A and B are unital C^*-algebras with states $\varphi \in S(A)$ and $\psi \in S(B)$, then for any C^*-norm $\|\cdot\|_\alpha$ on $A \odot B$, $\varphi \odot \psi$ extends to a state on $A \otimes_\alpha B$.

Proof. We will use the same method as in the last lemma, but we will require a bit of care. Fix a pure state $\varphi \in S(A)$, and let $\mathcal{S}_\varphi = \left\{\psi \in S(B) : \left|\varphi \odot \psi(x)\right| \leq \|x\|_\alpha \text{ for all } x \in A \odot B\right\}$.

We start by showing that for every self-adjoint $h \in B$, we have

$$\|h\| = \sup_{\psi \in \mathcal{S}_\varphi} \left|\psi(h)\right|.$$

Choose a pure state ψ on $C = C^*(H, 1_B)$ with $\left|\psi(h)\right| = \|h\|$. We may extend $\varphi \odot \psi$ to a state on $A \otimes_\alpha C \subseteq A \otimes_\alpha B$. We may then extend this to a state ξ on $A \otimes_\alpha B$. Since $\xi|_A = \varphi$ is pure, we have that ξ is an extension of $\varphi \otimes \xi|_B$, so we have that $\mathcal{S}_\varphi = S(B)$ as follows from the previous lemma.

Now, fix a state $\psi \in S(B)$, and let $\mathcal{S}_\psi = \left\{\varphi \in S(A) : \left|\varphi \odot \psi(x)\right| \leq \|x\|_\alpha \text{ for all } x \in A \odot B\right\}$. The previous paragraph gives that $\mathcal{S}_\psi$ contains all the pure states of A, so since it is also a convex w^*-closed subset of $S(A)$, we have that $\mathcal{S}_\psi = S(A)$ by the Krein–Milman theorem. □

This allows us to prove Takesaki's Theorem.

Theorem (Takesaki's Theorem)**:** Let A and B be arbitrary C^*-algebras. Then, $\|\cdot\|_{\min}$ is the smallest C^*-norm on $A \odot B$.

Proof. We start by assuming that A and B are unital and separable. We may find faithful states $\varphi \in S(A)$ and $\psi \in S(B)$. If $\|\cdot\|_\alpha$ is any C^*-norm on $A \odot B$, then by the previous lemma we may extend $\varphi \odot \psi$ to a state $\varphi \otimes_\alpha \psi$. By the uniqueness of GNS representations, we have that the representations $\pi_{\varphi \otimes_\alpha \psi}$ restricted to $A \odot B$ and $\pi_\varphi \odot \pi_\psi$ are unitarily equivalent. Since we chose faithful states, both π_φ and π_ψ are faithful representations, so the norm closure of $\pi_\varphi \odot \pi_\psi(A \odot B)$ is isomorphic to $A \otimes B$. Therefore, $A \otimes B$ is isomorphic to $\pi_{\varphi \otimes_\alpha \psi}(A \otimes_\alpha B)$. Since this is a quotient of $A \otimes_\alpha B$, we have that $\|\cdot\|_{\min} \leq \|\cdot\|_\alpha$.

For the non-unital case, we may extend any C^*-norm to the unitizations, and obtain the same result. □

Corollary: For any A and B, and any C^*-norm $\|\cdot\|_\alpha$ on $A \odot B$, there are natural surjective $*$-homomorphisms

$$A \otimes_{\max} B \to A \otimes_\alpha B \to A \otimes_{\min} B.$$

Lemma: If $\|\cdot\|_\alpha$ is a C^*-norm on $A \odot B$, then it is a cross norm.

Proof. Let $A \otimes_\alpha B \subseteq B(H)$. It suffices to show that $\|a \otimes b\| = 1$ for any positive norm-one elements $a \in A$ and $b \in B$, as we may use the C^*-identity to establish the rest of the desired results.

Let $\varepsilon > 0$, and let $e = \chi_{[1-\varepsilon,1]}(a)$ and $f = \chi_{[1-\varepsilon,1]}(b)$ be some spectral projections. Notice that e and f commute, since they are elements of the WOT closures of the ranges of the restriction maps to $A \otimes_\alpha B$. We claim that $ef \neq 0$.

For this, observe that there are nonzero elements $a_0 \in C^*(a)$ and $b_0 \in C^*(b)$ such that $a_0 \leq e$ and $b_0 \leq f$, so that $0 \neq a_0 \otimes b_0 \leq ef$.

Therefore, for a unit vector $\xi \in efH$, we have $\|(a \otimes b)\xi - \xi\| \leq 2\varepsilon$. Since ε was arbitrary, it follows that $\|a \otimes b\| = 1$. □

Ultraproducts

Let $\{M_n\}_{n\in\mathbb{N}}$ is a family of finite von Neumann algebras with respective traces τ_n, we let $\omega \in \beta\mathbb{N}$ be a non-principal ultrafilter. Define

$$\ell_\infty(\mathbb{N}, (M_n)_n) = \left\{ (x_n)_n \in \prod_{n\in\mathbb{N}} M_n : \sup_{n\in\mathbb{N}} \|x_n\| < \infty \right\}.$$

This is also known as the ℓ_∞-direct product of the family $\{M_n\}_{n\in\mathbb{N}}$. The *trace ideal* is given by

$$I = \left\{ (x_n)_n \in \ell_\infty(\mathbb{N}, (M_n)_n) : \lim_{n\to\omega} \tau_n(x_n^* x_n) = 0 \right\}.$$

Proposition: The set I is a closed two-sided ideal in $\ell_\infty(\mathbb{N}, (M_n)_n)$.

Proof. Let $(x_n)_n \in I$. Then, for any $a_n, b_n \in M_n$, we have

$$\|a_n x_n b_n\|_\tau \leq \|a_n\|\|b_n\|\|x_n\|_\tau.$$

Therefore, we have

$$\begin{aligned} \lim_{n\to\omega} \tau_n(b_n^* x_n^* a_n^* a_n x_n b_n) &\leq \lim_{n\to\omega} (\|a_n\|\|b_n\|\|x_n\|_\tau) \\ &= 0. \end{aligned}$$

□

Definition: If $\{M_n\}_{n\in\mathbb{N}}$ is a family of finite von Neumann algebras with respective traces τ_n, then the algebra $\ell_\infty(\mathbb{N},(M_n)_n)/I$ is known as the *ultraproduct* of the family $\{M_n\}_{n\in\mathbb{N}}$, equipped with norm

$$\left\|(x_n)_n\right\| = \lim_{n\to\omega}\|x_n\|.$$

If $M_n = M$ is fixed, we call this the *ultrapower* of M, and it is denoted by M^ω. The von Neumann algebra M admits a diagonal embedding into the ultrapower M^ω, which will still be denoted by M.

Theorem: The ultraproduct of a family of finite von Neumann algebras is a finite von Neumann algebra with trace $\tau_\omega := \lim_{n\to\omega}\tau_n$. The ultraproduct is a factor whenever each of the M_n is a factor.

Definition: Let M be a tracial von Neumann algebra. We say M has property Γ if $M'\cap M^\omega \neq \mathbb{C}$ for some non-principal ultrafilter ω on $\mathbb{N}$. We say M is McDuff if $M'\cap M^\omega$ is non-abelian.

One of the main advantages of ultrapowers in von Neumann algebra theory is that they convert asymptotic behavior in a von Neumann algebra to a definite property. That is, approximation type properties in von Neumann algebras correspond directly to definite properties in the ultrapower, and vice versa.

In particular, this means that there are asymptotic definitions for property Γ and the McDuff property as described below.

Definition: Let M be a tracial von Neumann algebra. We say M has property Γ if there exists a sequence of unitaries $(u_n)_{n\in\mathbb{N}}$ with $\tau(u_n) = 0$ and

$$\lim_{n\to\infty}\|u_n x - x u_n\|_\tau = 0$$

for every $x \in M$.

We say M is McDuff if $M \cong M \otimes R$, where R denotes the hyperfinite II_1 factor.

One of the main questions concerning ultrapowers is whether every II_1 factor admits an embedding into R^ω for any ultrapower of R. This was shown to be false in the general case (via a result from quantum complexity theory), but this question is still open for the case of (infinite conjugacy class) group von Neumann algebras.

Crossed Products

If we have a group Γ acting on a C^*-algebra A, then a question becomes how we may construct a C^*-algebra that encodes this action.

If a group is acting on another group, there is a construction known as the (outer) semidirect product that enables this. We will construct a C^*-algebra emerging from an action $\alpha\colon \Gamma \to \mathrm{Aut}(A)$, which will be dubbed $A \rtimes_\alpha \Gamma$. If $\Gamma \curvearrowright A$, then we call A a Γ-C^*-algebra. The triple (A,Γ,α) is called a C^*-dynamical system.

If A is a Γ-C^*-algebra, then we let $C_c(\Gamma, A)$ denote the space of finitely supported functions on Γ with values in A. We equip $C_c(\Gamma,A)$ with a α-twisted product and $*$-operation by taking, for $S = \sum_{s\in\Gamma} a_s s$ and $T = \sum_{t\in\Gamma} b_t t$ in $C_c(\Gamma, A)$,

$$\begin{aligned} S *_\alpha T &= \sum_{s,t\in\Gamma} a_s\alpha_s(b_t)st \\ S^* &= \sum_{s\in\Gamma}\alpha_{s^{-1}}(a_s^*)s^{-1}. \end{aligned}$$

We note that this definition emerges from the fact that we are turning $C_c(\Gamma,A)$ into a $*$-algebra where the action of Γ is an inner automorphism.

Note that if $A = \mathbb{C}$ and α is the trivial action, then this construction recovers the group ring $\mathbb{C}[\Gamma]$.

To turn this into a C^*-algebra, we must consider two different C^*-norms.

A *covariant representation* $(u,\pi,\mathcal{H})$ of a Γ-C^*-algebra A given by a unitary representation $(u,\mathcal{H})$ of Γ and a $*$-representation $(\pi,\mathcal{H})$ of A that satisfies

$$u_s\pi(a)u_s^* = \pi(\alpha_s(a)).$$

Every covariant representation gives rise to a $*$-representation of $C_c(\Gamma,A)$ (by extending linearly), and every nondegenerate $*$-representation arises in this fashion. Denote by $u\times\pi$ the associated $*$-representation of $C_c(\Gamma,A)$.

Definition: The *full crossed product* of a C^*-dynamical system is the completion of $C_c(\Gamma,A)$ with respect to the norm

$$\|x\|_u = \sup\{\|\pi(x)\| : \pi \text{ is a cyclic } *\text{-homomorphism into } B(\mathcal{H})\}.$$

We denote this object by $A\rtimes_\alpha\Gamma$.

It is not apparent at first that this is a norm. Yet, this crossed product yields a universal property.

Proposition: Let $(u,\pi,\mathcal{H})$ be a covariant representation of a Γ-C^*-algebra A. Then, there is a $*$-homomorphism $\sigma\colon A\rtimes_\alpha\Gamma\to B(\mathcal{H})$ such that

$$\sigma\left(\sum_{s\in\Gamma}a_s s\right) = \sum_{s\in\Gamma}\pi(a_s)u_s$$

for all $\sum_{s\in\Gamma}a_s s\in C_c(\Gamma,A)$.

We will now define the reduced crossed product for a Γ-C^*-algebra A. Start with a faithful representation of $A\subseteq B(\mathcal{H})$. Then, we may define a representation of A on $\mathcal{H}\otimes\ell_2(\Gamma)$ by

$$\pi(a)\left(v\otimes\delta_g\right) = \left(\alpha_{g^{-1}}(a)(v)\right)\otimes\delta_g,$$

where δ_g is the orthonormal basis for $\ell_2(\Gamma)$. This is effectively the direct sum representation

$$\begin{aligned}\pi(a) &= \bigoplus_{g\in\Gamma}\alpha_g^{-1}(a)\\ &\in B\left(\bigoplus_{g\in\Gamma}\mathcal{H}\right).\end{aligned}$$

This allows us to see that the left-regular representation of Γ spatially implements the action α. That is,

$$\begin{aligned}(1\otimes\lambda_s)\pi(a)(1\otimes\lambda_s^*)\left(v\otimes\delta_g\right) &= (1\otimes\lambda_s)\pi(a)\left(v\otimes\delta_{s^{-1}g}\right)\\ &= (1\otimes\lambda_s)\left(\left(\alpha_{g^{-1}s}(a)(v)\right)\otimes\delta_{s^{-1}g}\right)\\ &= \left(\alpha_{g^{-1}s}(a)(v)\right)\otimes\delta_g\\ &= \left(\alpha_{g^{-1}}(\alpha_s(a))(v)\right)\otimes\delta_g\\ &= \pi(\alpha_s(a))\left(v\otimes\delta_g\right).\end{aligned}$$

This gives us the covariant representation $(1\otimes\lambda)\times\pi$. This is called a regular representation.

Definition: The *reduced crossed product* of a C^*-dynamical system (A,Γ,α) is the norm closure of the image of some regular representation $C_c(\Gamma,A)\to B(\mathcal{H}\otimes\ell_2(\Gamma))$.

The fact that this crossed product is independent of the faithful representation is analogous to the proof that the spatial norm on the tensor product is independent of the choice of faithful representations for the tensor factors.

Proposition: The reduced crossed product $A\rtimes_{\alpha,r}\Gamma$ does not depend on the choice of faithful representation for A.

Proof. Let $F \subseteq \Gamma$ be finite, and let $P \in B(\ell_2(\Gamma))$ be the finite rank projection onto the span of $\left\{\delta_g : g \in F\right\}$. Let $\left\{e_{p,q}\right\}_{p,q\in F}$ be the matrix units of $PB(\ell_2(\Gamma))P \cong \mathbb{M}_F(\mathbb{C})$, and fix elements $a \in A$ and $s \in \Gamma$. Let $\pi\colon A \to B(\mathcal{H}\otimes\ell_2(\Gamma))$ be a regular representation. Our first claim is that

$$\begin{aligned}(1\otimes P)\pi(a) &= (1\otimes P)\pi(a)(1\otimes P)\\ &= \sum_{q\in F}\alpha_q^{-1}(a)\otimes e_{q,q}.\end{aligned}$$

This follows from the fact that we may view $\pi(a)$ as

$$\pi(a) = \sum_{q\in\Gamma}\alpha_q^{-1}(a)\otimes e_{q,q},$$

an SOT-convergent sum. In particular, we have

$$\begin{aligned}(1\otimes P)\pi(a)(1\otimes\lambda_s)(1\otimes P) &= \left(\sum_{q\in F}\alpha_q^{-1}(a)\otimes e_{q,q}\right)(1\otimes P\lambda_s P)\\ &= \left(\sum_{q\in F}\alpha_q^{-1}(a)\otimes e_{q,q}\right)\left(\sum_{p\in F\cap sF} 1\otimes e_{p,s^{-1}p}\right)\\ &= \sum_{p\in F\cap sF}\alpha_p^{-1}(a)\otimes e_{p,s^{-1}p}\\ &\in A\otimes\mathbb{M}_F(\mathbb{C}).\end{aligned}$$

If we select some element of $C_c(\Gamma, A) \subseteq B(\mathcal{H}\otimes\ell_2(\Gamma))$ given by

$$x = \sum_{s\in\Gamma} a_s\lambda_s,$$

then

$$\begin{aligned}(1\otimes P)x(1\otimes P) &= \sum_{s\in\Gamma}\sum_{p\in F\cap sF}\alpha_p^{-1}(a_s)\otimes e_{p,s^{-1}p}\\ &\in A\otimes\mathbb{M}_F(\mathbb{C}).\end{aligned}$$

Since $A\otimes\mathbb{M}_F(\mathbb{C})$ has a unique norm, it follows that the norm of $(1\otimes P)x(1\otimes P)$ does not depend on the embedding of A into $B(\mathcal{H})$. ☐

Corollary: An element $x = \sum_{s\in\Gamma} a_s\lambda_s \in C_c(\Gamma, A)$ is positive in $A \rtimes_{\alpha,r} \Gamma$ if and only if for any finite sequence $s_1,\dots,s_n \in \Gamma$, the operator $\left(\alpha_{s_i}^{-1}\left(a_{s_i s_j^{-1}}\right)\right)_{i,j} \in \mathbb{M}_n(A)$ is positive.

Proof. An operator is positive if and only if its compression by any projection is positive, so we have

$$\begin{aligned}(1\otimes P)x(1\otimes P) &= \sum_{s\in\Gamma}\sum_{p\in F\cap sF}\alpha_p^{-1}(a_s)\otimes e_{p,s^{-1}p}\\ &\in A\otimes\mathbb{M}_F(\mathbb{C})\end{aligned}$$

can be identified with the operator matrix $\left(\alpha_{s_i}^{-1}\left(a_{s_i s_j^{-1}}\right)\right)_{i,j} \in \mathbb{M}_n(A)$ if we let $F = \{s_1,\dots,s_n\}$ by setting $p = s_i$ and $s_j = s^{-1}p$. ☐

There is a C^*-dynamical version of Fell's Absorption Principle.

Proposition: If $(u, \mathrm{id}_A, \mathcal{H})$ is a covariant representation, then the covariant representation

$$(u \otimes \lambda, \mathrm{id}_A \otimes 1, \mathcal{H} \otimes \ell_2(\Gamma))$$

is unitarily equivalent to a regular representation. There is a natural $*$-isomorphism

$$C^*((u \otimes \lambda)(\Gamma), A \otimes 1) \cong A \rtimes_{\alpha,r} \Gamma.$$

Proof. Define a unitary U on $\mathcal{H} \otimes \ell_2(\Gamma)$ by taking $U(\xi \otimes \delta_t) = u_t\xi \otimes \delta_t$.

Then,

$$\begin{aligned} U(1 \otimes \lambda_s)U^* &= (u_s \otimes \lambda_s) \\ U\left(\sum_{t \in G} \alpha_t^{-1}(a) \otimes e_{t,t}\right)U^* &= a \otimes 1 \end{aligned}$$

for all $s \in \Gamma$ and $a \in A$. □

Now, we will discuss the existence of conditional expectations from the crossed product $A \rtimes_{\alpha,r} \Gamma$ onto A.

Lemma: Let ψ be a faithful state on B. Then, $\mathrm{id}_A \otimes \psi \colon A \otimes B \to A$ is faithful.

Proof. We observe that the set

$$\{f \otimes g : f \in A^*, g \in B^*\} \subseteq (A \otimes B)^*$$

separates the points of $A \otimes B$. Since $A \otimes B \subseteq B(\mathcal{H} \otimes \mathcal{K})$, we have that the vector states arising from elementary tensors $h \otimes k$ separate the points of $B(\mathcal{H} \otimes \mathcal{K})$.

Now, if $x \in (A \otimes B)_+$ is nonzero, there is a state φ on A such that $(\varphi \otimes \mathrm{id}_B)(x) \in B$ is nonzero and positive. Since ψ is faithful, it follows that

$$\begin{aligned} 0 &< \psi((\varphi \otimes \mathrm{id}_B)(x)) \\ &= \varphi((\mathrm{id}_A \otimes \psi)(x)), \end{aligned}$$

so that $(\mathrm{id}_A \otimes \psi)(x)$ is nonzero. □

Proposition: The map

$$\begin{aligned} E \colon C_c(\Gamma, A) &\to A \\ \sum_s a_s \lambda_s &\mapsto a_e \end{aligned}$$

extends to a faithful conditional expectation from $A \rtimes_{\alpha,r} \Gamma$ onto A. In particular,

$$\max_{s \in \Gamma} \|a_s\|_A \leq \left\| \sum_{s \in \Gamma} a_s \lambda_s \right\|_{A \rtimes \Gamma}.$$

Proof. Let $(u, \mathrm{id}_A, \mathcal{H})$ be a covariant representation. By Fell's absorption principle, we may view $A \rtimes_{\alpha,r} \Gamma$ as a C^*-subalgebra of $B(\mathcal{H}) \otimes C^*_\lambda(\Gamma)$.

In this representation, the map E is the restriction of $\mathrm{id}_{B(\mathcal{H})} \otimes \tau$ to the crossed product, where τ is the canonical state on $C^*_\lambda(\Gamma)$. Therefore, E is faithful.

Note that $a_s = E(z\lambda_s^*)$, where $z = \sum_s a_s \lambda_s$, which gives the desired inequality. □

Observe that E is Γ-equivariant, in that

$$E\left(\lambda_s z \lambda_s^{-1}\right) = \alpha_s(E(z))$$

for every $s \in \Gamma$ and $z \in A \rtimes_{\alpha,r} \Gamma$.

Actions by $\mathbb{Z}$

Suppose $A \subseteq B(\mathcal{H})$ and $\alpha \in \operatorname{Aut}(A)$ is an automorphism. We can use α to denote the induced action of $\mathbb{Z}$ given by $n \mapsto \alpha^n$. We will let $[k,n] = \{k, k+1, \dots, n\}$ be the interval of integers between k and n.

The finite subsets $F_n = [0,n]$ form a Følner sequence for $\mathbb{Z}$; for each k, we have

$$\begin{aligned} \frac{|(k+F_n)\cap F_n|}{|F_N|} &= \frac{|[k,n+k]\cap[0,n]|}{n+1} \\ &= \frac{n-k+1}{n+1} \\ &\to 1. \end{aligned}$$

We now recall a lemma from [Pau02].

Lemma: Let A be a C^*-algebra. Every positive element in $\mathbb{M}_n(A)$ is a sum of n elements of the form $\left(a_i^* a_j\right)_{i,j}$.

Lemma: If A is a Γ-C^*-algebra, and $F \subseteq \Gamma$ is a finite set, then for each set $\left\{a_p\right\}_{p\in F} \subseteq A$, we have that the element

$$\sum_{p,q\in F} \alpha_p\left(a_p^* a_q\right)\lambda_{pq^{-1}} \in C_c(\Gamma, A)$$

is positive.

Proof. The element in question is equal to

$$\left(\sum_{p\in F} a_p \lambda_{p^{-1}}\right)^*\left(\sum_{p\in F} a_p \lambda_{p^{-1}}\right).$$

□

Lemma: If A is a Γ-C^*-algebra, and $F \subseteq \Gamma$ is a finite set, then there exist completely positive contractions $\varphi\colon A \rtimes_{\alpha,r} \Gamma \to A \otimes \mathbb{M}_F(\mathbb{C})$ and $\psi\colon A \otimes \mathbb{M}_F(\mathbb{C}) \to C_c(\Gamma) \subseteq A \rtimes_{\alpha,r} \Gamma$ such that for all $a \in A$ and $s \in \Gamma$, we have

$$\psi \circ \varphi(a\lambda_s) = \frac{|F \cap sF|}{|F|} a\lambda_s.$$

Proof. We already know that there is a completely positive contraction $\varphi\colon A \rtimes_{\alpha,r} \Gamma \to A \otimes \mathbb{M}_F(\mathbb{C})$ such that

$$\varphi(a\lambda_s) = \sum_{p\in F\cap sF} \alpha_p^{-1}(a) \otimes e_{p,s^{-1}p}.$$

We use the fact that if $\alpha\colon \Gamma \to \operatorname{Aut}(A)$ is an action, and $\tau{\otimes}\alpha\colon \Gamma \to \operatorname{Aut}(B\otimes A)$ is defined by $(\tau\otimes\alpha)_g = \mathrm{id}_B \otimes \alpha_g$, then the crossed products

$$(B\otimes A) \rtimes_{\tau\otimes\alpha,r} \Gamma \cong B \otimes (A \rtimes_{\alpha,r} \Gamma)$$

are isomorphic. Specifically, we apply this result to the case of $B = \mathbb{M}_F(\mathbb{C})$.

It suffices to prove that

$$\begin{aligned} \psi\colon A \otimes \mathbb{M}_F(\mathbb{C}) &\to C_c(\Gamma, A) \subseteq A \rtimes_{\alpha,r} \Gamma \\ a \otimes e_{p,q} &\mapsto \frac{1}{|F|}\alpha_p(a)\lambda_{pq^{-1}} \end{aligned}$$

is positive. Yet, by the characterization of positivity in $\mathbb{M}_n(A)$, this means we only need to verify for every set $\left\{a_p\right\}_{p\in F} \subseteq A$, we have

$$\psi\left(\sum_{p,q\in F} a_p^* a_q \otimes e_{p,q}\right) \geq 0.$$

This gives

$$\psi\left(\sum_{p,q\in F} a_p^* a_q \otimes e_{p,q}\right) = \sum_{p,q\in F} \frac{1}{|F|}\alpha_p\left(a_p^* a_q\right)\lambda_{pq^{-1}},$$

but this element is positive. $\square$